Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Flood disasters in Central Vietnam repeatedly sever road infrastructure, rendering conventional relief network models – which assume full network connectivity – fundamentally inadequate. We formulate the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), a two-stage stochastic program that jointly minimizes expected logistics cost and expected maximum deprivation cost over a set of flood scenarios, while explicitly accounting for broken inter-hub links, multi-modal transport, and pre-positioned inventory. Deprivation cost is modeled as an exponential penalty on victim waiting time, and Daganzo's continuous approximation is embedded for tractable last-mile routing cost estimation. To solve MO-IHLNDP, we propose PB-NSGA (Priority-Based Nondominated Sorting Genetic Algorithm), which confines the chromosome to first-stage hub decisions and reconstructs all scenario-dependent second-stage variables through a structured priority-based decoder, reducing search-space complexity from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}|)$. Computational experiments on AP and TR81 hub location benchmarks confirm near-optimal recovery against exact Pareto fronts on verifiable instances and sub-quadratic runtime scaling. A case study across four flood-prone provinces in Central Vietnam yields a methodological framework for decision support: a quantified cost-equity trade-off curve for evidence-based resource allocation, and a demonstration of how the model – when supplied with empirical data – can identify robust hub cores and guide transport mode diversification to preserve network connectivity under extreme disaster scenarios.

Keywords: Humanitarian logistics · Hub location · Multi-objective optimization · Stochastic programming · Disaster relief · Evolutionary algorithm

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1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [1]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [2].

There were many great challenges in humanitarian operations during this period. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains, while key arteries such as National Highway 1 were severed by landslides, isolating communities for days [2]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes "incomplete", characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing [3].

1.2 Related Works

Humanitarian logistics and facility location. Humanitarian logistics differs fundamentally from its commercial counterpart in that speed and equity dominate over cost minimisation [4]. A comprehensive review of humanitarian facility location under uncertainty [5] identifies stochastic programming and robust optimisation as the dominant paradigms, while Boonmee et al. [6] document the growing need for models that jointly capture disruption, uncertainty, and equity. To account for social welfare, Holguín-Veras et al. [7] introduced *deprivation cost* – an exponential penalty on victim waiting time.

Hub location and incomplete networks. Hub-and-spoke structures consolidate flows efficiently and have been studied extensively since Campbell's seminal formulations [8]. For disaster settings, the *incomplete* hub network – in which infrastructure damage severs inter-hub arcs – is a more realistic topology than the classical fully-connected graph [9]. Sangsawang and Chanta [10] proposed a pioneering flood-specific hub location model, a single-objective deterministic formulation for Thailand, establishing that hub-and-spoke structures yield measurable gains in relief distribution efficiency. Multi-modal hub networks further address the heterogeneous transport modes (road, waterway, air) that characterise disaster response [11].

Stochastic and robust optimisation for relief networks. The two-stage stochastic programming framework naturally separates pre-disaster strategic decisions from post-disaster recourse [12]. Under demand and supply uncertainty, robust formulations [13] and prepositioning models have been shown to outperform deterministic baselines. Disruption of transportation infrastructure compounds this uncertainty by rendering pre-planned routes infeasible [14], a condition endemic to Central Vietnam's flood disasters.

Multi-objective evolutionary algorithms. NSGA-II [15] is the standard framework for multi-objective combinatorial problems in humanitarian logistics, owing to its parameter-free Pareto ranking and elitist selection. Zhou et al. [16] applied a multi-objective evolutionary algorithm to multi-period dynamic emergency resource scheduling under uncertain road networks, demonstrating the effectiveness of population-based search when scenario-dependent decisions are coupled to shared strategic variables. Li et al. [11] designed a bi-objective multimodal hub-and-spoke network for emergency relief using a meta-heuristic approach, showing that problem-specific customized operators are essential for maintaining feasibility across complex network topologies.

Research gap and contributions. No existing work simultaneously addresses incomplete hub network design, two-stage stochastic programming, multi-objective humanitarian objectives (logistics cost and deprivation cost), and a flood-disaster case study. Sangsawang and Chanta [10] is the closest precedent, yet it is deterministic and single-objective. This paper closes the gap by formulating the MO-IHLNDP and solving it via PB-NSGA, a priority-based [17] evolutionary algorithm.

2 Problem Description and Mathematical Model

We model the network design problem as a capacitated, single-allocation MO-IHLNDP. The network comprises **origins** (supply sources), **demands** (victim groups), and **hubs** of two types: planned (pre-disaster) and reactive (post-disaster, dynamically opened).

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption − damaged roads and buildings − can disable pre-programmed plans, urging the need for a dynamic solution [13]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [12]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that SAR operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

The model integrates the deprivation cost [7] for social equity, and Daganzo's continuous approximation (CA) [18] to tractably estimate last-mile rescue times.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F^a_{ks}	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u, and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i, and relief items provided at origin j in scenario s
c_{jks}	Pre-computed cheapest transport cost from origin j to hub k in scenario s: $c_{jks} = \min_{m \in \mathcal{M} \mid a_{jkms}=1} C_{jkm}$, ($+\infty$ if no available mode)
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuity factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s, 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s, 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s, 0 otherwise
$q_k \geq 0$	Amount of relief items inventoried at hub k
$f_{khms} \geq 0$	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$
\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k\in\mathcal{H}} F_k x_k + \sum_{k\in\mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s\in\mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k\in\mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j\in\mathcal{J}} \sum_{k\in\mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\
& + \underbrace{\sum_{k\in\mathcal{H}} \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k\in\mathcal{H}} \sum_{i\in\mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg]
\end{aligned}
\tag{1}
$$

Objective Z_1 minimizes total expected logistics cost. Using the previously mentioned CA [18], demands are aggregated on a grid basis to pre-compute the last-mile routing cost Θ_{kis}:

$$
\Theta_{kis} = \min_{m\in\mathcal{M}} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)
$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$
\text{Min } Z_2 = \sum_{s\in\mathcal{S}} \pi_s \left(\max_{i\in\mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is} \cdot \Omega_{is}} - 1 \right) \right] \right) \tag{2}
$$

Objective Z_2 minimizes the expected maximum deprivation cost. Applying the exponential penalty on waiting times (Ω_{is}) ensures equitable resource distribution across all populations.

$$
\Omega_{is} = \sum_{k\in\mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m\in\mathcal{M}} \tau_{kim} \right)
$$

The sensitivity coefficient λ_{is} is proportional to the risk index r_{is} ($\lambda_{is} = \lambda_0(1 + r_{is})$), effectively prioritizing vulnerable populations. Although Z_2 introduces non-linearity, our meta-heuristic solution approach can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property ($\sum_{k\in\mathcal{H}} z_{iks} = 1$).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$
\begin{aligned}
\text{Minimize} \quad & Z_1, Z_2 \\
\text{subject to} \quad & \text{Constraints (3)} - \text{(16)}
\end{aligned}
$$

$$
x_k + y_{ks} \le 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{3}
$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{4}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{5}$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{6}$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \tag{7}$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{8}$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{9}$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{10}$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{11}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \tag{12}$$

$$\begin{aligned} \sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} \leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\ + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \tag{13}$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{14}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{15}$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \tag{16}$$

Constraints (3) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (4), while the reactive hubs are required to be placed within safe zones (5). Constraints (6)-(11) rigorously enforce the single-allocation property to active hubs via available links. Constraint (12) restricts flow to be trans-shipped on intact paths. Constraints (13) and (14) maintain flow conservation and capacity limits. Specifically, (13) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (15)-(16).

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [19]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [15] with a custom heuristic decoder.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0,1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0,1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \ldots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i. The continuous vector $\mathbf{W} \in [0,1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency $(\lambda \cdot D)$ and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$. If none qualify, the safest hub is forced active.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score prioritising urgency, isolation, and proximity to active hubs, enhanced with Gaussian noise to prevent premature convergence. Demands are then processed in descending score order.

Step 3 – Tiered Hub Assignment. For each demand i, we trial active hubs in proximity order from the anchor hub π_{A_i}. *Pass 1* evaluates candidates based on residual capacity, travel time, and operational status. If unsuccessful, *Pass 2* relaxes capacity constraints, or eventually opens the nearest safe inactive hub as a reactive fallback. Infeasibility accumulates as constraint violation.

Step 4 – Supply Routing and Objective Accumulation. Origins supply the most under-supplied active hubs, and surplus is redistributed via discounted inter-hub trans-shipment (α). Expected objectives Z_1 and Z_2 are accumulated across all $s \in \mathcal{S}$. Algorithm 1 formalises these procedures.

3.3 Genetic Operators and Selection

Selection uses binary tournament on Pareto rank with crowding distance as tiebreaker. *Crossover* ($p_c = 0.9$) applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX ($\eta_c = 20$) to $\mathbf{R}$ and $\mathbf{W}$. *Mutation* ($p_m = 1/|\mathbf{C}|$) applies Bit-flip to $\mathbf{X}$, random-reset to $\mathbf{A}$, and Polynomial mutation ($\eta_m = 20$) to $\mathbf{R}$ and $\mathbf{W}$. Constraint handling follows the constrained-dominance principle [15]: feasible solutions always dominate infeasible ones; among infeasible solutions, lower CV dominates. When a partial Pareto front must be split, tiebreaking proceeds by (1) Pareto rank, (2) crowding distance, (3) minimum Hamming distance in $\mathbf{X}$-space – preserving genotypic diversity in hub configuration and countering premature convergence.

Algorithm 1 Priority-Based Decoder (single scenario s)

Require: Chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$, scenario s, instance data
Ensure: $(Z_1^s,\ Z_2^s,\ \mathrm{CV}^s)$

```
Activate $\mathcal{H}_{\text{act}} \leftarrow \{k : X_k = 1,\ r_{ks} \le \chi\}$; set $q_k \leftarrow R_k \kappa_k$
if $\mathcal{H}_{\text{act}} = \emptyset$ then
    Force-activate $\arg\min_k r_{ks}$ among $\{k : X_k = 1\}$
end if
Score demands by urgency, isolation, and proximity; sort descending
for each demand $i$ (in priority order) do
    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$; trial order $\leftarrow \pi_{A_i}$
    Pass 1: $k^* \leftarrow \arg\max \mathrm{HubScore}(k)$ over first $K$ active, reachable hubs with residual $> 0$
    if Pass 1 succeeds then
        Assign $i \rightarrow k^*$; deduct demand from residual inventory of $k^*$
    else if any active reachable hub remains then
        Pass 2: assign $i$ to first such hub (capacity violation allowed)
    else
        Reactive fallback: open nearest safe inactive hub; add $F^a_{ks}$; $\mathrm{CV}^s \mathrel{+}= 1$
    end if
end for
Assign origins greedily; redistribute surplus via inter-hub trans-shipment (discount $\alpha$)
Accumulate $Z_1^s$, $Z_2^s$
return $(Z_1^s,\ Z_2^s,\ \mathrm{CV}^s)$
```

4 Computational Experiments

4.1 Experimental Setup

PB-NSGA is implemented in C++17; dataset construction and post-processing are performed in Python 3, on a workstation with an Intel Core i7-12700 (2.1 GHz, 12 cores) and 16 GB RAM. Fixed for all runs: $N = 100$, $G = 200$, $p_c = 0.9$, $\eta_c = \eta_m = 20$, $p_m = 1/|\mathbf{C}|$, decoder noise $\sigma = 0.05$. Problem constants are $\chi = 0.7$, $\alpha = 0.6$, $\gamma = 3.0$ kg/person, $\lambda_0 = 0.8$. Each configuration is executed for 20 independent runs, reporting mean ± std. We use **Hypervolume (HV)** [20] (reference point $(1.1, 1.1)$) and **IGD$^+$** [21] for performance metrics, utilizing exact Pareto fronts from complete enumeration as the ground truth for small instances.

4.2 Dataset Description

Benchmark instances. We adapt two standard hub location datasets: the **AP** set [22] ($n \in \{10, 20, 25, 40, 50, 100\}$) and the real inter-city **TR81** dataset [23] (81 Turkish provincial capitals). Each is transformed into a DRND instance by assigning hub/origin/demand roles and generating three flood scenarios (mild, severe, extreme) with probabilities $(0.60, 0.30, 0.10)$. Three transport modes (truck, motorboat, helicopter) with varying capacities (60, 25, 10 persons) and speeds are employed. Roads are subject to flood disruption, while air/water modes remain available.

Central Vietnam case study. We construct two instances modelling four flood-prone provinces: Da Nang, Quang Nam, Thua Thien-Hue, and Quang Ngai (Figure 1). **CV-Small** uses district-level demand nodes ($|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$); **CV-Large** uses commune-level resolution ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$). Node coordinates are drawn from real administrative centroids; hub candidates correspond to confirmed logistics facilities and rescue stations; supply origins include seaports, river terminals, and highland border nodes. Scenario risk indices and road disruption probabilities are calibrated from a flood vulnerability score combining topographic, hydrological, and coastal exposure criteria [24].

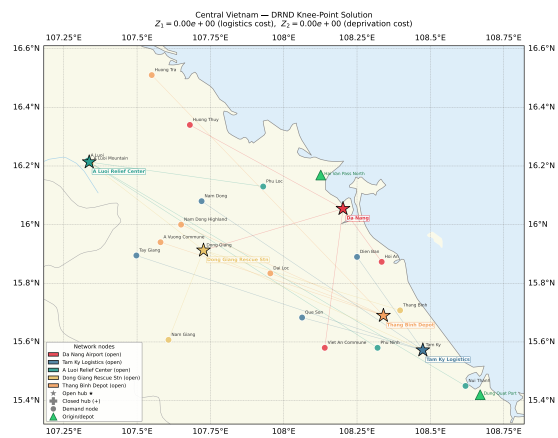

Fig. 1. Central Vietnam study region with demand nodes (circles), hub candidates (squares), and supply origins (triangles). Node shading encodes static vulnerability r_u^a; the four flood-prone provinces (Da Nang, Quang Nam, Thua Thien-Hue, Quang Ngai) constitute the study region.

4.3 Experiment 1: Benchmarking on HLP Instances

For instances with $|\mathcal{H}| \leq 12$, exact Pareto fronts are obtained via branch-and-bound exhaustive enumeration (30-minute limit), providing IGD$^+$ ground truth for AP10–AP40. Table 1 reports results across all benchmark instances.

PB-NSGA achieves near-zero IGD$^+$ against the exact front on AP10–AP40, confirming that the priority-based decoder consistently recovers near-optimal hub configurations on verifiable instances. On AP40 ($|\mathcal{H}| = 8$), the mean IGD$^+$ of 0.015 ± 0.010 indicates solutions within 1.5% of the true Pareto front on average. Runtime scales sub-linearly from AP10 (0.31 s) to AP100 (2.20 s), well within operational planning horizons; Figure 2 shows the full scaling profile.

Table 1. PB-NSGA on HLP benchmarks (mean ± std, 20 runs). †: IGD^+ against exact Pareto front from complete enumeration.

Instance ($\|\mathcal{H}\|$)	HV (norm.)	IGD^+	#Pareto	Time (s)
AP10 (3)†	1.113 ± 0.092	0.063 ± 0.060	10.2 ± 22.9	0.31 ± 0.02
AP20 (4)†	1.073 ± 0.160	0.116 ± 0.150	4.6 ± 2.2	0.47 ± 0.02
AP25 (5)†	1.155 ± 0.102	0.052 ± 0.091	6.2 ± 1.7	0.55 ± 0.03
AP40 (8)†	1.190 ± 0.014	0.015 ± 0.010	7.6 ± 6.7	0.82 ± 0.05
AP50 (10)	1.179 ± 0.023	0.043 ± 0.029	15.2 ± 20.3	1.01 ± 0.11
AP100(20)	1.088 ± 0.058	0.067 ± 0.045	17.9 ± 22.9	2.20 ± 0.19
TR81 (12)	1.061 ± 0.063	0.092 ± 0.055	5.8 ± 3.1	1.54 ± 0.13

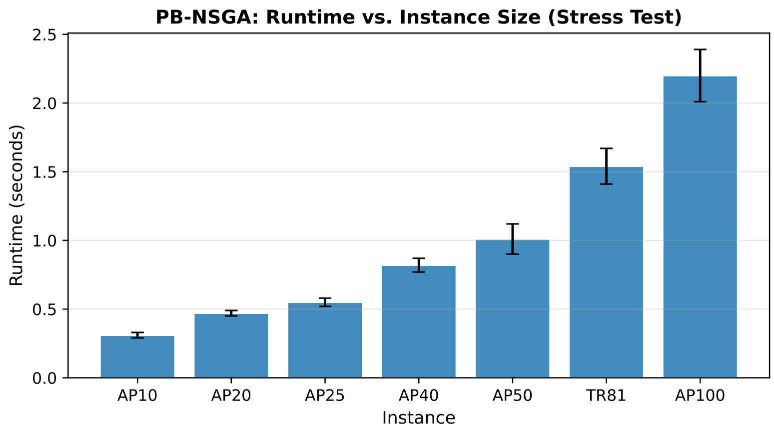

Fig. 2. PB-NSGA mean CPU time (s) per benchmark instance (20 seeds). Error bars show ±1 standard deviation. Runtime grows sub-linearly across the full instance range.

4.4 Experiment 2: Case Study – Central Vietnam

Table 2 reports PB-NSGA performance on both Central Vietnam instances; Figure 3 shows the combined non-dominated fronts pooled across 20 seeds.

The fronts exhibit a convex cost-deprivation trade-off: marginal reductions in expected maximum deprivation Z_2 require disproportionately large increases in logistics cost Z_1 toward the low-cost end. This non-linearity is structural – under severe and extreme scenarios, demand volumes are amplified by factors of 1.8 and 2.8 respectively, so a network optimised for Z_1 alone provides insufficient coverage in the tail events that govern Z_2.

Figure 4 maps the three representative solutions across disaster scenarios, illustrating how the active hub set and transport routes reorganise as conditions escalate. Figure 5 further quantifies scenario sensitivity through a hub × scenario risk heatmap ordered by Pareto-front selection frequency.

Table 2. PB-NSGA on Central Vietnam instances (mean ± std, 20 runs).

Instance	HV (norm.)	IGD$^+$	#Pareto
CV-Small ($\|\mathcal{I}\|$=20, $\|\mathcal{H}\|$=5)	1.164 ± 0.100	0.029 ± 0.064	6.0 ± 3.1
CV-Large ($\|\mathcal{I}\|$=100, $\|\mathcal{H}\|$=20)	1.037 ± 0.058	0.064 ± 0.025	17.1 ± 20.2

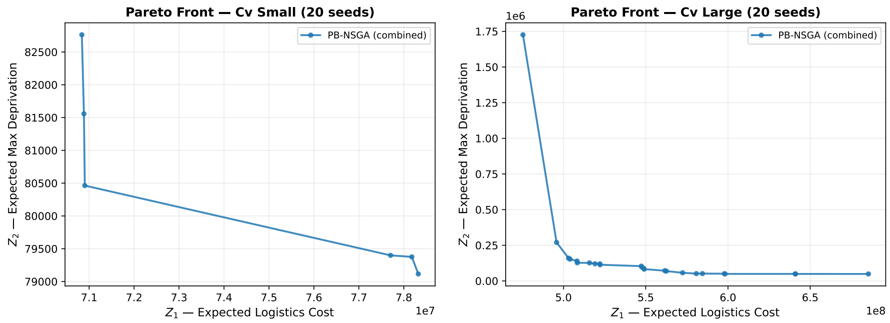

Fig. 3. Combined Pareto fronts for CV-Small (left) and CV-Large (right), pooled across all seeds. Three representative solutions – minimum cost (⧫), balanced (★), and minimum deprivation (•) – are annotated.

Under the mild scenario, a compact set of coastal hubs served primarily by truck provides adequate coverage. As the event intensifies to extreme, breached hubs force demand to reroute through inland sites via motorboat and helicopter. A deterministic single-scenario model would suppress this adaptation, systematically under-investing in resilient transport capacity.

Sample Average Approximation and Out-Of-Sample Validation To address potential overfitting to the three training scenarios, we further validate the CV-Large Pareto front (17 solutions from Seed 0) against two rigorous numerical experiments. First, we generated a Sample Average Approximation (SAA) set of 100 scenarios, incorporating continuous Gaussian risk decay and stochastic epicenter placements. Evaluating the predetermined first-stage hub configurations and decoder Weights $\mathbf{W}$ on this SAA set resulted in 0.00 mean constraint violations (CV), confirming that the network maintains strict feasibility across a dense distribution of disaster realizations. Expected logistics cost (Z_1) and deprivation (Z_2) understandably scaled by +20.8% and +143.7% on average, reflecting the inclusion of multiple severe tail-events in the 100-scenario envelope compared to the baseline.

Second, we introduced an Out-Of-Sample (OOS) "Double Typhoon" scenario – a novel extreme event featuring four simultaneous epicenters and a severity multiplier of 3.5×, strictly outside the training parameters. Remarkably, the heuristic decoder successfully routed and processed demand for all 17 Pareto solutions with exactly 0 CV. This proves that the priority-based variables (es-

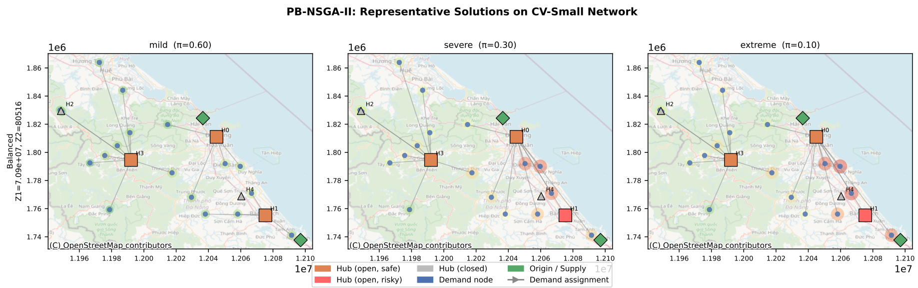

Fig. 4. Representative PB-NSGA solutions for CV-Small across mild (left), severe (centre), and extreme (right) scenarios. Open squares denote active hubs; arrows indicate demand–hub assignment; hatched squares mark hubs exceeding safety threshold $\chi = 0.7$.

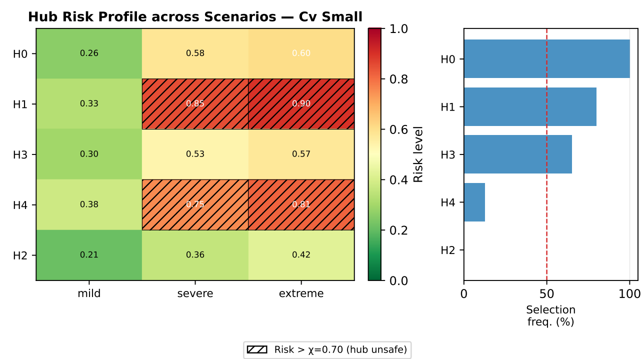

Fig. 5. Hub risk profile across scenarios for CV-Small. Rows are ordered by Pareto-front selection frequency (right panel, %); hatched cells indicate risk above $\chi = 0.7$. Hubs frequently selected yet unsafe in the extreme scenario define the critical resilience gap.

pecially the hub-choice anchors $\mathbf{A}$ and priority weights $\mathbf{W}$) learn a generalized emergency response policy, rather than merely overfitting to the specific geometries of the training set.

Methodological Framework for Decision Support *Robust hub prioritisation.* Assuming the input risk parameters accurately reflect topological vulnerability, hubs with high Pareto-front selection frequency and consistently low risk across all scenarios would constitute a robust core warranting permanent infrastructure investment. Hubs that are cost-effective under mild conditions but exceed χ under extreme events define a *resilience gap*: identifying these gaps allows planners to maintain contingency routing or backup inventory at adjacent low-risk sites.

Transport mode diversification. The extreme scenario forces a modal shift away from road transport precisely when the network is most degraded. Co-locating coastal hubs with waterway terminals and maintaining helicopter access at a subset of sites provides asymmetric resilience benefit at a fraction of the cost of additional hub capacity.

Quantified cost of resilience. The balanced Pareto solution requires only a modest logistics cost premium over the minimum-cost solution, yet substantially reduces expected maximum deprivation under severe and extreme scenarios. This trade-off curve provides humanitarian agencies with a quantitative tool for communicating resource needs to funders.

5 Conclusion and Future Work

We formulated the MO-IHLNDP to balance logistics costs against social equity under infrastructure disruptions, solving it with the proposed PB-NSGA. Demonstrating its utility on a Central Vietnam case study, the framework identifies robust hub configurations, quantifies cost-equity trade-offs, and guides modal diversification for flood resilience.

Limitations. The absence of a publicly available real-world relief dataset for Central Vietnam required the use of a synthetic instance calibrated from administrative and geographic data. Field validation against actual disaster records remains necessary before operational deployment.

Future work will prioritize three directions. First, constructing a real relief logistics dataset for Central Vietnam in collaboration with local authorities, to enable rigorous empirical validation. Second, rapid practical deployment by integrating the model with government relief planning systems and validating against historical flood response data. Third, strengthening PB-NSGA through improved decoder strategies, adaptive operator tuning, and hybridization with local search to enhance solution quality on large-scale instances.

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